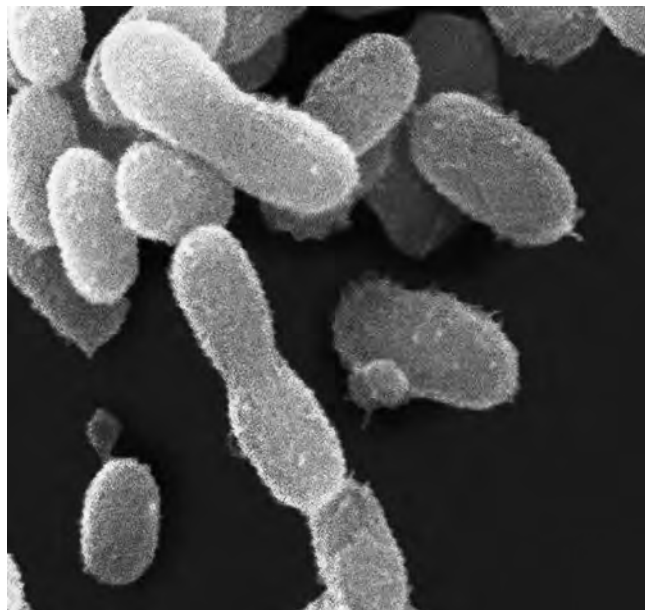


- 5 A new type of bacteria was discovered in Greenland in 2008. It had survived for more than 120 000 years in the ice of a glacier.

The photograph shows this new type of bacteria, as seen using an electron microscope.



(Source: © PENN STATE UNIVERSITY/SCIENCE PHOTO LIBRARY)

This type of bacteria can survive in low temperatures and low oxygen levels.

- (a) Describe how scientists could determine that this was a new type of bacteria.

(2)

.....

.....

.....

.....

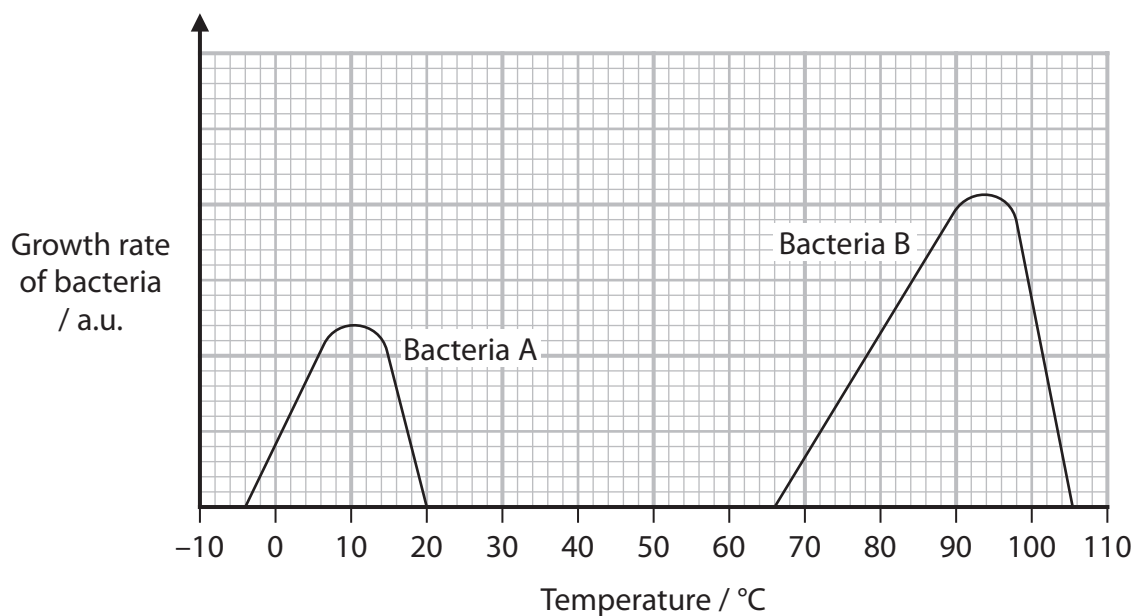
.....

.....

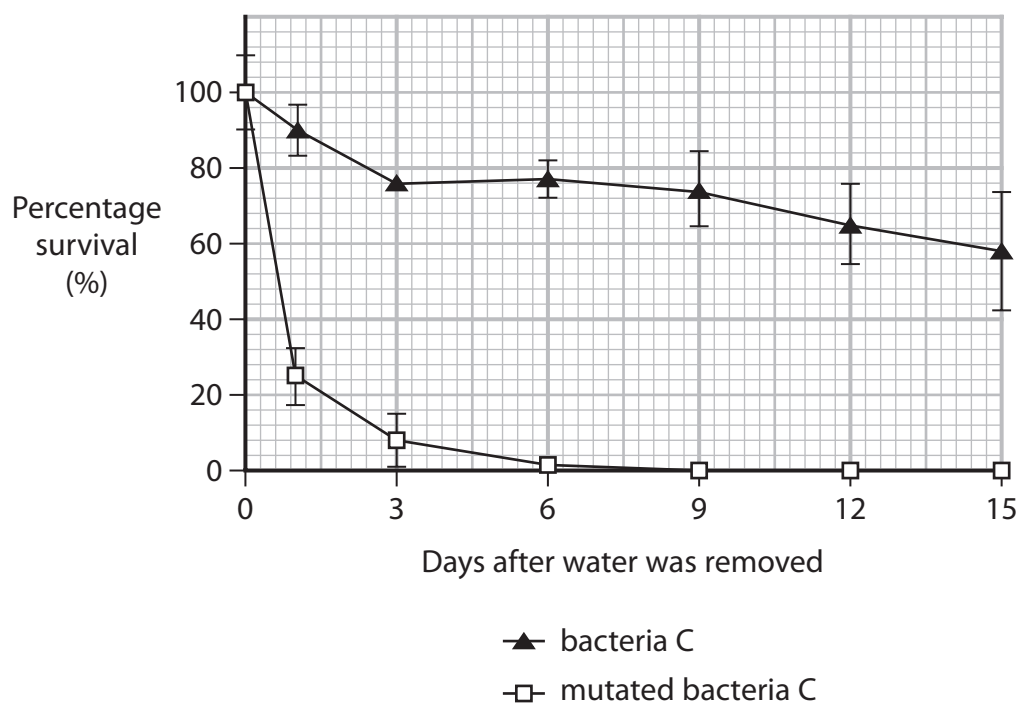
*(b) Bacterial growth and survival can be affected by changing conditions.

The effect of temperature and desiccation (water removal) on different bacteria was investigated.

Graph 1 shows the effect of temperature on the growth rate of bacteria A and bacteria B.



Graph 2 shows the percentage survival of bacteria C and mutated bacteria C after water was removed from the surroundings.



Explain the changes in growth rate and percentage survival of these bacteria.

Use the information in the graphs and your own knowledge to support your answer.

(6)

(Total for Question 5 = 8 marks)

